

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	JWT-based user registration and login with salted password hashing (passlib + bcrypt). Each user's data is isolated and protected.	High
2	Authorization levels	Single user role (Borrower) with full access to own loans, settlements, and AI history. Admin role planned for future versions.	Medium

3	External interfaces	Google Gemini AI API for generating personalized negotiation letters and strategies; FastAPI REST endpoints for all CRUD operations.	High
4	Transaction processing	User submits financial profile and loan data via HTTP POST; backend calculates health metrics, priority scores, and settlement recommendations in real-time. AI letters generated via Gemini API with fallback template logic.	High
5	Reporting & History	Dashboard displays aggregated financial statistics (total outstanding, EMI ratio, stress level, health score). Users can view settlement history and AI-generated letter history.	High
6	Business rules	EMI-to-income ratio classification: <30% = Low stress, 30-50% = Medium, >50% = High. Settlement percentage based on stress level: High = 40%, Medium = 60%, Low = 75%. Overdue months >6 reduces settlement % by 10. Priority scoring formula: $(\text{overdue_months} \times 10) + (\text{interest_rate} \times 2) + \min(\text{outstanding}/10000, 20) + \min(\text{EMI}/1000, 10)$.	High

7	Compliance	User passwords hashed using bcrypt; JWT tokens for session management; no sensitive financial data exposed in API responses. Complies with general data privacy principles.	High
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